

3

Simplicial complexes

The reader may have noticed that although surfaces are constructed out of triangles I did not define the word ‘triangle’ in Chapter 2. Doubtless the reader has come across triangles before. Nevertheless a precise definition in terms suited to our purpose is given below, where triangles appear under the alias of ‘2-simplexes’. The precise definition makes it clear that ‘triangle’ is a good way to continue the sequence ‘point, segment, ...’ (which becomes ‘0-simplex, 1-simplex, ...’) and suggests that the fourth term should be ‘solid tetrahedron’: this is our three-dimensional building block. We shall need building blocks in higher dimensions too, but most examples in the text will be, as hitherto, two-dimensional. Algebraic concepts closely analogous to the cycles of Chapter 1 will be introduced in Chapter 4.

It is possible, by judicious skipping in this chapter, to avoid contact with simplexes of dimension higher than two. All initial definitions are stated for the case of two dimensions as well as in general; from 3.9 just assume $n \leq 2$.

Everything in the chapter takes place in a real vector space $\mathbb{R}^N$. Elements of $\mathbb{R}^N$ are called *points* or *vectors*. Unless otherwise stated the only restriction on N is that it should be large enough for the discussion to make sense; thus if we speak of four non-coplanar points then obviously N must be at least 3.

Simplexes

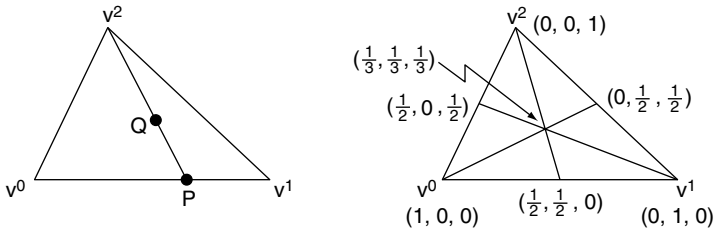
It is clear that two points v^0 and v^1 are the end-points of a segment if and only if they are distinct. The segment then consists of all points $(1 - \lambda)v^0 + \lambda v^1$ where $0 \leq \lambda \leq 1$, i.e. it is the set of points

$$\{\lambda_0 v^0 + \lambda_1 v^1 : \lambda_0 + \lambda_1 = 1, \quad \lambda_0 \geq 0, \quad \lambda_1 \geq 0\}.$$

Similarly three points v^0, v^1, v^2 are the vertices of a triangle if and only if they are not collinear, i.e. if and only if $v^1 - v^0$ and $v^2 - v^0$ are linearly independent. In the left-hand diagram

$$P = \lambda v^0 + (1 - \lambda)v^1 \quad (0 \leq \lambda \leq 1),$$

$$Q = \mu(\lambda v^0 + (1 - \lambda)v^1) + (1 - \mu)v^2 \quad (0 \leq \mu \leq 1).$$



Hence

$$Q = \lambda_0 v^0 + \lambda_1 v^1 + \lambda_2 v^2$$

where $\lambda_0 = \lambda\mu$, $\lambda_1 = (1 - \lambda)\mu$ and $\lambda_2 = 1 - \mu$. Notice that the λ_i are ≥ 0 and have sum 1. Conversely given non-negative $\lambda_0, \lambda_1, \lambda_2$ with sum 1 we can solve for λ and μ ; hence the triangle is the set of points

$$\{\lambda_0 v^0 + \lambda_1 v^1 + \lambda_2 v^2 : \lambda_0 + \lambda_1 + \lambda_2 = 1, \lambda_i \geq 0 \text{ for } i = 1, 2, 3\}.$$

The advantage of the λ_i over λ and μ is that any given point Q has uniquely determined λ_i (compare 3.5) whereas v^2 has $\mu = 0$ but arbitrary λ . The numbers $\lambda_0, \lambda_1, \lambda_2$ are called the *barycentric coordinates* of Q . The barycentric coordinates of some points in the triangle are given in the right-hand diagram above.

It can be shown that if we drop the condition of non-negativeness on the λ_i the above set of points is enlarged to include all those in the plane of the triangle. Points with all $\lambda_i > 0$ are inside the triangle, points with two $\lambda_i > 0$ and one zero are on an edge, and points with two λ_i equal to zero are at a vertex.

Definition 3.1. Let $v^0, \dots, v^n$ be $n + 1$ vectors in $\mathbb{R}^N$, $n \geq 1$. They are called *affinely independent* (*a-independent*) if $v^1 - v^0, \dots, v^n - v^0$ are linearly independent. By convention if $n = 0$ then the vector v^0 is always *a-independent* (even if it is zero).

Examples 3.2. (1) v^0 and v^1 are a -independent if and only if $v^0 \neq v^1$; v^0, v^1, v^2 are a -independent if and only if they are not collinear.

(2X) Let $v^0, \dots, v^n$ be in $\mathbb{R}^N$. They are a -independent if and only if, for all real numbers $\lambda_0, \dots, \lambda_n$, we have

$$\begin{pmatrix} \lambda_0 v^0 + \dots + \lambda_n v^n = 0 \\ \lambda_0 + \dots + \lambda_n = 0 \end{pmatrix} \Rightarrow \lambda_0 = \dots = \lambda_n = 0.$$

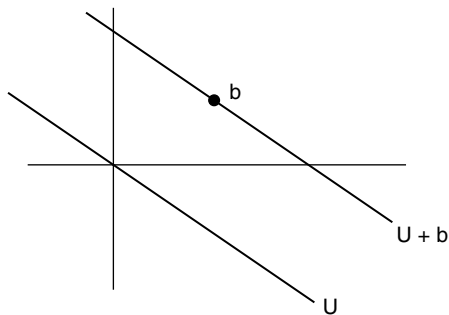
[Hint: $\lambda_0 v^0 + \dots + \lambda_n v^n = 0 = \lambda_1(v^1 - v^0) + \dots + \lambda_n(v^n - v^0)$ when $\sum \lambda_i = 0$.] This makes it clear that the definition of a -independence does not in reality depend on the ordering of the v^i : we could equally well define it by the linear independence of $v^0 - v^1, v^2 - v^1, v^3 - v^1, \dots, v^n - v^1$.

Also, adding a constant to all the v^i does not affect their a -independence or otherwise. (In contrast, even if v and w are linearly independent, $v - v$ and $w - v$ are not.) Thus a -independence does not depend on the choice of origin.

★(3) Let U be a subspace of $\mathbb{R}^N$ and let $b \in \mathbb{R}^N$, the set of vectors

$$U + b = \{u + b : u \in U\}$$

is called an *affine subspace* of $\mathbb{R}^N$ of dimension equal to that of U . We say that $U + b$ is *parallel* to U (it certainly has no points in common with U provided $b \notin U$, and if $b \in U$ then $U + b = U$).



For example, let $N = 2$ and let U be a line through the origin. Then $U + b$ is the line parallel to U , through b .

Notice that, given $U + b$, we can recover U :

$$U = \{v - w : v, w \in U + b\}.$$

On the other hand b is not uniquely determined by $U + b$ since in fact

$$U + b = U + b' \Leftrightarrow b - b' \in U.$$

(For the example above, these say that we can recover U by moving $U + b$ parallel to itself until it passes through the origin; and that replacing b by any $b' \in U + b$ gives the same line parallel to U .)

If $v^0, \dots, v^n$ are a -independent then $v^1 - v^0, \dots, v^n - v^0$, being linearly independent, are a basis for a subspace U of $\mathbb{R}^N$ of dimension n . The affine subspace $U + v^0$ then has dimension n and contains all of $v^0, \dots, v^n$. Notice that any affine subspace $U' + b$ containing $v^0, \dots, v^n$ must have dimension $\geq n$, for U' will contain the n linearly independent vectors $v^1 - v^0, \dots, v^n - v^0$. Hence the smallest affine subspace containing $v^0, \dots, v^n$ has dimension n .

Definition 3.1 can be reformulated: $v^0, \dots, v^n$ are a -independent if and only if the smallest affine subspace of $\mathbb{R}^N$ containing them has dimension n .

★

Definition 3.3. Let $v^0, \dots, v^n$ be vectors in $\mathbb{R}^N$. A vector v is said to be *affinely dependent* (a -dependent) on them if there exist real numbers $\lambda_0, \dots, \lambda_n$ such that

$$\lambda_0 + \dots + \lambda_n = 1 \quad \text{and} \quad v = \lambda_0 v^0 + \dots + \lambda_n v^n.$$

Examples 3.4. (1X) If $v^0, \dots, v^n$ are a -independent then none of them is a -dependent on the rest. If v is a -dependent on $v^0, \dots, v^n$ then $v^0, \dots, v^n, v$ are not a -independent.

★(2) The set of vectors a -dependent on $v^0, \dots, v^n$ is the smallest affine subspace of $\mathbb{R}^N$ containing $v^0, \dots, v^n$ and is called the *affine span* of $v^0, \dots, v^n$. ★

Proposition 3.5. Let $v^0, \dots, v^n$ be a -independent and let v be a -dependent on them. Then there exist unique real numbers $\lambda_0, \dots, \lambda_n$ such that $\sum \lambda_i = 1$ and $v = \sum \lambda_i v^i$. The λ 's are called the *barycentric coordinates* of v with respect to $v^0, \dots, v^n$.

Proof The λ 's exist by 3.3. To prove uniqueness suppose that $\sum \lambda_i v^i = \sum \mu_i v^i$ where $\sum \lambda_i = \sum \mu_i = 1$ (all sums run from 0 to n). Then $\sum v_i v^i = 0$, where $v_i = \lambda_i - \mu_i$ so that $\sum v_i = 0$. Hence by 3.2(2) we have $v^0 = \dots = v^n = 0$, i.e. $\lambda_i = \mu_i$ for $i = 0, \dots, n$. □

By analogy with the discussion preceding 3.1 we can now define the general concept of simplex.

Definition 3.6. Let $v^0, \dots, v^n$ be a -independent. The (closed) *simplex* with vertices $v^0, \dots, v^n$ is the set of points a -dependent on $v^0, \dots, v^n$ and with every barycentric coordinate ≥ 0 . The simplex is also said to be *spanned*

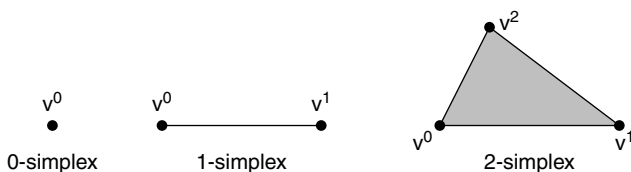
by $v^0, \dots, v^n$. The points with all barycentric coordinates > 0 are said to be *interior* to the simplex, and the set of interior points is sometimes called the *open simplex* with vertices $v^0, \dots, v^n$. The *boundary* of the simplex consists of those points which are not interior, i.e. which have at least one barycentric coordinate $= 0$.

Notation 3.7. In this book, the unqualified word ‘simplex’ will mean ‘closed simplex’. The simplex with a -independent vertices $v^0, \dots, v^n$ will be denoted by $(v^0 \dots v^n)$ and often represented by a symbol like s_n or just s . The integer n is called the *dimension* of $(v^0 \dots v^n)$ (compare 3.2(3)). A simplex of dimension n is called an n -simplex. The boundary of s_n will be denoted by $\dot{s}_n$ and the interior by $\overset{\circ}{s}_n$.

Examples 3.8. (1) For any vector v^0 the 0-simplex (v^0) is just the set $\{v^0\}$. We often denote (v^0) by merely v^0 . The open 0-simplex with vertex v^0 is also $\{v^0\}$. The boundary of (v^0) is empty.

For any two distinct vectors v^0, v^1 the 1-simplex $(v^0 v^1)$ is the closed *segment* with end-points v^0, v^1 . The boundary of $(v^0 v^1)$ is $\{v^0, v^1\}$, and the interior is the open segment with end-points v^0, v^1 .

For any three non-collinear vectors v^0, v^1, v^2 the 2-simplex $(v^0 v^1 v^2)$ is the *triangle* with vertices v^0, v^1, v^2 . The boundary of $(v^0 v^1 v^2)$ is the set of points on the edges of the triangle, i.e. $(v^0 v^1) \cup (v^1 v^2) \cup (v^2 v^0)$, and the interior is the rest of $(v^0 v^1 v^2)$.



For any four non-coplanar vectors v^0, v^1, v^2, v^3 the 3-simplex $(v^0 v^1 v^2 v^3)$ is the *solid tetrahedron* with vertices v^0, v^1, v^2, v^3 . Its boundary consists of four triangles.

★(2) A closed simplex is a closed subset of $\mathbb{R}^N$ in the usual sense that it contains all its limit points. This can be seen by means of barycentric coordinates: the i th barycentric coordinate of each point of a sequence of points in the simplex will be a real number ≥ 0 , so that the i th barycentric coordinate of the limit point of a convergent sequence will also be ≥ 0 . The interior of a closed simplex is the corresponding open simplex, which is not an open subset of $\mathbb{R}^N$ unless $n = N$. ★

(3X) A simplex determines its vertices – that is, if $(v^0 \dots v^n) = (w^0 \dots w^m)$ then $m = n$ and the v 's are merely a permutation of the w 's. To prove this we must find a characterization of the vertices of a simplex purely in terms of the set of points in the simplex. (This is perhaps a slightly unfamiliar situation – in euclidean geometry a triangle is defined in terms of its sides, and the sides intersect in pairs at the vertices, which are therefore determined. Here, on the other hand, a triangle is defined by the set of points inside it and on its boundary.) Such a characterization, which can be established using barycentric coordinates, is this: a point $v \in (v^0 \dots v^n)$ is a vertex (i.e. is one of the points $v^0, \dots, v^n$) if and only if it is not the mid-point of any segment joining two distinct points of $(v^0 \dots v^n)$.

(4X) A simplex is *convex*, that is if v and w are points of the simplex then every point of the (straight) segment joining v and w belongs to the simplex. Indeed the simplex is the smallest convex set containing its vertices, which is expressed by saying that it is the *convex hull* of its vertices.

(5X) Let $w \in (v^0 \dots v^n)$ and let $v \neq w$ be an interior point of $(v^0 \dots v^n)$. Show that the segment from v to w , produced beyond w if necessary, meets the boundary of the simplex in precisely one point. (This amounts to showing that there is precisely one point of the simplex of the form $(1 - \alpha)v + \alpha w$ with $\alpha \geq 1$ and having at least one barycentric coordinate zero.)

(6X) Let v be an interior point of the simplex $(v^0 \dots v^n)$. Show that $v^0, \dots, v^{n-1}, v$ are a -independent.

Definition 3.9. Let $s_n = (v^0 \dots v^n)$ be a simplex. A *face* of s_n is a simplex whose vertices form a (non-empty) subset of $\{v^0, \dots, v^n\}$. If the subset is proper (i.e. not the whole of $\{v^0, \dots, v^n\}$) we say that the face is a *proper face*. If s_p is a face of s_n we write $s_p < s_n$ or $s_n > s_p$; note that $s_n < s_n$ for any simplex s_n .

Remarks and Examples 3.10. (1) The boundary of s_n (3.6) is the union of the proper faces of s_n .

(2) A face of s_n consists of those points of s_n for which the barycentric coordinates attached to vertices not in the face are zero. It follows at once that two faces of s_n either are disjoint or else intersect in a face of s_n .

(3) Strictly speaking a vertex v^i of s_n is not a face of s_n . What is a face is the 0-simplex $(v^i) = \{v^i\}$. It does little harm, however, to fail to distinguish between v^i and (v^i) .

(4X) An n -simplex has $\binom{n+1}{p+1}$ faces of dimension p and $2^{n+1} - 1$ faces altogether.

(5) The zero- and one-dimensional faces of an n -simplex form a complete graph on $n + 1$ vertices.

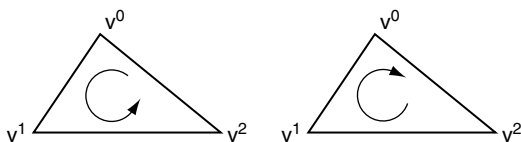
Ordered simplexes and oriented simplexes

A fairly detailed treatment of orientation is given in an appendix to this chapter, which is intended to motivate Definition 3.11 below and relate it to the concept of orientation in vector spaces. It is not necessary to read the appendix in order to follow this section. Orientations of 2-simplexes have already been introduced informally in connexion with closed surfaces (see 2.3). The algebraic theory developed from Chapter 4 onwards depends strongly on the concept of orientation, which is used to define a ‘boundary’ homomorphism analogous to that introduced in 1.19 for oriented graphs. (Compare 1.21(3).)

According to Definition 3.6, permutation of the vertices of a simplex leaves the simplex unchanged. If on the other hand we insist on a definite ordering of the vertices the result is called an *ordered simplex*. Thus associated with any n -simplex there are $(n + 1)!$ distinct ordered simplexes. In practice we do not wish to distinguish between all of these.

Consider for example a 2-simplex $(v^0 v^1 v^2)$. The six ordered simplexes associated with it are those in which the ordering is as follows:

$$\begin{array}{ll} (v^0 v^1 v^2) & (v^0 v^2 v^1) \\ (v^1 v^2 v^0) & (v^2 v^1 v^0) \\ (v^2 v^0 v^1) & (v^1 v^0 v^2) \end{array}$$



The three even permutations of (v^0, v^1, v^2) all order the vertices ‘anti-clockwise’ (taking this as the direction for (v^0, v^1, v^2) itself) while the three odd permutations order them ‘clockwise’. This suggests the following general definition.

Definition 3.11. Let $s_n = (v^0 \dots v^n)$ be an n -simplex. An *orientation* for s_n is a collection of orderings for the vertices consisting of a particular ordering and all even permutations of it. An *oriented n -simplex* σ_n is an n -simplex s_n together with an orientation for s_n .

Notation 3.12. We write $\sigma_n = (v^0 \dots v^n)$ to mean that σ_n is the oriented n -simplex with vertices $(v^0, \dots, v^n)$ and orientation given by the ordering displayed and all even permutations of it. It will always be clear from the context whether $(v^0 \dots v^n)$ means a simplex or an oriented simplex. When $n > 0$ we write $-\sigma_n$ for the *oriented simplex* consisting of the same n -simplex and the collection of other possible orderings of the vertices. Thus by definition $-(-\sigma_n) = \sigma_n$.

Remarks 3.13. (1) A 0-simplex has a unique orientation; an n -simplex for $n \geq 1$ has two orientations.

(2) A 1-simplex has two orderings and two orientations: ordering and orientation are the same in this case. They are represented diagrammatically by an arrow from the first to the second vertex (compare the diagrams following 1.3). An orientation of a triangle is indicated by a circular arrow as in the above diagrams.

(3) For any permutation π of $0, \dots, n$ the oriented simplexes $\sigma_n = (v^0 \dots v^n)$ and $\tau_n = (v^{\pi(0)} \dots v^{\pi(n)})$ satisfy $\sigma_n = \tau_n$ if π is even, and $\sigma_n = -\tau_n$ if π is odd.

Simplicial complexes

Simplicial complexes are the most general geometrical objects we study in this book. They include as special cases the graphs of Chapter 1 and the closed surfaces of Chapter 2, and they provide a good framework for developing an algebraic theory analogous to the cycles of Chapter 1. As before we impose ‘intersection conditions’ (compare 1.3, 2.1); now that we have the ideas of simplex and face at our disposal it is possible to state these very succinctly.

Definitions 3.14. A *simplicial complex* is a finite set K of simplexes in $\mathbb{R}^N$ with the following two properties:

- (1) if $s \in K$ and $t < s$ (t is a face of s (3.9)) then $t \in K$
- (2) intersection condition: if $s \in K$ and $t \in K$ then $s \cap t$ is either empty or else a face both of s and of t .

The *dimension* of K is the largest dimension of any simplex in K . The *underlying space* of K , denoted $|K|$, is the set of points in $\mathbb{R}^N$ which belong to at least one simplex of K . That is, $|K|$ is the union of the simplexes of K . Occasionally, if Σ is merely a set of simplexes we use $|\Sigma|$ to denote the union of the simplexes in Σ .

Notice that if $s \in K$ then all the faces of s are also elements of K . It may seem odd that we should explicitly include these faces in K since they are, as subsets of $\mathbb{R}^N$, contained in s : the faces of s do not contribute anything extra to the underlying space. The reason is to be found in the next chapter, where algebraic apparatus is introduced which requires the presence of all these faces. It is worth emphasizing at this point that a simplicial complex is a finite set of simplexes in $\mathbb{R}^N$ while the portion of $\mathbb{R}^N$ which these simplexes together occupy (i.e. the underlying space) is generally an infinite set of points in $\mathbb{R}^N$. (The underlying space $|K|$ is finite if and only if K has dimension 0.)

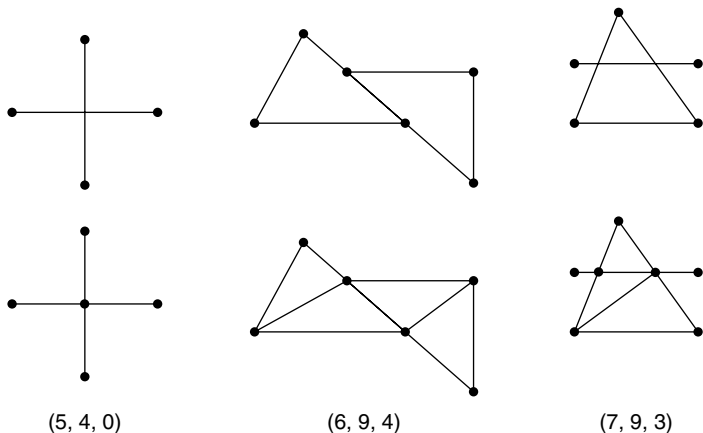
When *drawing simplicial complexes* it is sometimes helpful to indicate vertices by a heavy dot and to shade in 2-simplexes which are part of the complex. We shall occasionally make use of this in what follows. But for the majority of the diagrams we just use the convention that straight segments are 1-simplexes, intersections of segments (and no other points) are 0-simplexes, and all triangular regions in the diagram are in fact spanned by 2-simplexes. From now on the reader should assume this latter convention unless there is indication (usually by selective shading) to the contrary.

Examples 3.15. (1) A *graph* is a simplicial complex of dimension 1. This agrees with 1.3, although admittedly no explicit distinction is made there between the finite set of vertices and edges (the simplicial complex) and the, usually infinite, underlying space.

(2) The empty collection of simplexes is a simplicial complex whose underlying space is empty. This simplicial complex does not have a dimension, unless we make a special convention to cover it. A popular choice is -1 .

(3) A closed surface M as defined in 2.1 consists of triangles (i.e. 2-simplexes) satisfying three conditions one of which is an intersection condition. If we let M^+ be the set of triangles, edges and vertices in M then M^+ is a simplicial complex of dimension 2: thus a closed surface is a simplicial complex provided we explicitly mention the vertices and edges as well as the triangles. In the future we shall assume this has been done. A closed surface is then a special kind of simplicial complex called ‘pure’: a simplicial complex K of dimension n is called *pure* if every simplex of K is a face of some n -simplex of K . An example of a non-pure simplicial complex is $K = \{(v^0v^1), (v^0), (v^1), (v^2)\}$ where v^0, v^1, v^2 are distinct. Here $\dim K = 1$ but (v^2) is not a face of any 1-simplex of K .

(4X) The three diagrams below do not represent simplicial complexes (Why?), but below them are diagrams for genuine simplicial complexes with, in each case, the same underlying space. The triples in brackets are the numbers of 0, 1 and 2-simplexes respectively.



(5X) The set of all faces of an n -simplex s_n is a simplicial complex of dimension n (see 3.10(2)), which is denoted by $\bar{s}_n$. The set of all proper faces of s_n is a simplicial complex of dimension $n - 1$ (even for $n = 0$ if we adopt the convention of (2) above). It is denoted $\dot{s}_n$ and is called the *boundary* of s_n . Thus (compare 3.10(1)) the ‘boundary of s_n ’ refers either to the set of proper faces of s_n or to the union of the proper faces of s_n . This will not cause difficulty.

(6XX) The conditions (1) and (2) of 3.14 are equivalent to (1) and the following condition (2)’.

(2)’ If s and $t \in K$, where $s \neq t$, then the open simplexes $\overset{\circ}{s}$ and $\overset{\circ}{t}$ with the same vertices as s and t are disjoint.

It is not too hard to prove (1) & (2) $\Rightarrow$ (1) & (2)’. The proof of the converse implication is more difficult. One method is to use induction, assuming the result true whenever K contains only simplexes of dimension $< n$, and proving it when K contains only simplexes of dimension $\leq n$. The induction hypothesis then applies to the subset of K consisting of simplexes of dimension $< n$ and since open simplexes are disjoint we have $s \cap t = \overset{\circ}{s} \cap \overset{\circ}{t}$ for any $s, t \in K$. Hence by the induction hypothesis $s \cap t$ is a union of common faces of s and t . However it is also convex (see 3.8(4)) and it is not hard to show that a convex union of (closed) faces of a simplex is itself a face.

Definitions 3.16. (1) A *subcomplex* of a simplicial complex K is a subset L of K which is itself a simplicial complex. (This is so if and only if L has the property that

$$s \in L \quad \& \quad t < s \Rightarrow t \in L,$$

since 3.14(2) is automatically satisfied.) The subcomplex L is called *proper* if $L \neq K$.

(2) An *oriented simplicial complex* is a simplicial complex in which every simplex is provided with an orientation.

Thus every simplicial complex can certainly be oriented, usually in many different ways. This is not to be confused with the coherent orientation of a closed surface, as in 2.3, which is not always possible.

By orienting a simplicial complex each subcomplex becomes oriented in a natural way. Orientations of 1-simplexes and 2-simplexes will be indicated by arrows and circular arrows respectively.

Examples 3.17. (1X) If K_1 and K_2 are subcomplexes of K then so are $K_1 \cup K_2$ and $K_1 \cap K_2$. How about $K_1 \setminus K_2$?

(2XX) Let K_1 and K_2 be simplicial complexes in $\mathbb{R}^N$ and suppose that $|K_1| \cap |K_2| = |L|$, where L is a subcomplex both of K_1 and of K_2 . Show that $K_1 \cup K_2$ is a simplicial complex. (There is no difficulty showing that (1) of 3.14 holds, and the only tricky case of (2) is when $s \in K_1$ and $t \in K_2$. It is probably easiest to prove that, supposing $s \neq t$, the open simplexes determined by s and t are disjoint, and then to use 3.15(6).)

(3) Let r be an integer ≥ 0 . The r -*skeleton* of a simplicial complex K is the set of simplexes of K with dimension $\leq r$; it is denoted K^r . Clearly K^r is a subcomplex of K . The 1-skeleton K^1 of any simplicial complex of K is a graph.

(4) A simplicial complex K is called *connected* if K^1 is a connected graph, i.e. if there is a path along the edges of K^1 connecting any two given vertices. For any K , a *component* of K is a maximal connected subcomplex; thus K is the (disjoint) union of its components, and K is connected if and only if it has exactly one component.

Abstract simplicial complexes and realizations

A simplicial complex K is determined by (i) its set V of vertices and (ii) the subsets of V which span simplexes of K . This information can be presented abstractly, and the result is an ‘abstract simplicial complex’ which can be ‘realized’ to give K . The definitions are parallel to those given for graphs in Chapter 1, and again there is a theorem that realization is always possible. We shall make use of these ideas in the next section, in an attempt to clarify the meaning of certain diagrams representing simplicial complexes.

Definition 3.18. An *abstract simplicial complex* is a pair $X = (V, S)$ where V is a finite set whose elements are called the *vertices* of X and S is a set of non-empty subsets of V . Each element $s \in S$ is called a *simplex* of X , and if $s \in S$ has precisely $n + 1$ elements ($n \geq 0$), s is called an n -simplex. (Thus an ‘abstract’ simplex is merely the set of its vertices.) S is required to satisfy the following two conditions:

- (1) $v \in V \Rightarrow \{v\} \in S$
- (2) $s \in S, t \subset s, t \neq \emptyset \Rightarrow t \in S$.

The *dimension* of X is, so long as $V \neq \emptyset$, the largest n for which S contains an n -simplex. (The dimension of $(\emptyset, \emptyset)$ is defined to be -1 .)

Two abstract simplicial complexes $X = (V, S)$ and $X' = (V', S')$ are said to be *isomorphic* if there is a bijection $f : V \rightarrow V'$ with the property that

$$\{v^0, \dots, v^n\} \in S \Leftrightarrow \{f(v^0), \dots, f(v^n)\} \in S'.$$

It is clear that any simplicial complex K determines an abstract simplicial complex $X = (V, S)$: take V to be the set of vertices of K and let $\{v^0, \dots, v^n\} \in S$ if and only if $(v^0 \dots v^n) \in K$. Two simplicial complexes are called *isomorphic* if the abstract simplicial complexes they determine are isomorphic. A *realization* of an abstract simplicial complex X is a simplicial complex K whose corresponding abstract simplicial complex is isomorphic to X . Thus the vertices of K can be so labelled that a set of vertices of K spans an n -simplex of K if and only if the corresponding set of vertices of X is an n -simplex of X .

This idea generalizes 1.3 and the following theorem generalises 1.4. Readers interested only in two-dimensional simplicial complexes should substitute $n = 2$ throughout the theorem and proof. The proof could in any case safely be omitted on a first reading.

Theorem 3.19. *Every abstract simplicial complex of dimension n has a realization in $\mathbb{R}^{2n+1}$. (Note that any two realizations of the same abstract simplicial complex are by definition isomorphic.)*

Sketch of Proof Consider the ‘rational normal curve’

$$C = \{(x, x^2, \dots, x^{2n+1}) \in \mathbb{R}^{2n+1} : x \in \mathbb{R}\}.$$

I claim that any $2n + 2$ distinct points of C are a -independent. The proof is similar to that in 1.4 of the case $n = 1$: it amounts to checking that a certain determinant (a Vandermonde determinant) does not vanish.

Now let $X = (V, S)$ be an abstract simplicial complex and take points $P_1, \dots, P_k$ on C , one for each element of V ($k = |V|$). If $s \in S$ then $s \subset V$ and $|s| \leq n + 1 < 2n + 2$ so the P 's corresponding to elements of s are a -independent and span a $(|s| - 1)$ -simplex. Let K be the set of such simplexes for all $s \in S$. It remains to prove that K is a simplicial complex for, if it is, then the underlying abstract simplicial complex is certainly X again.

Take two simplexes in K and suppose that their sets of vertices overlap. Then we can take them to be

$$(P_1 \dots P_\alpha \dots P_\beta) \text{ and } (P_\alpha \dots P_\beta \dots P_\gamma)$$

say, where $\gamma < 2n + 2$ since $\dim X = n$, so that $P_1, \dots, P_\gamma$ are a -independent. Using the definition of a -independence and 3.2(2) it is not hard to show that these simplexes intersect precisely in $(P_\alpha \dots P_\beta)$ which is a common face. If the simplexes have disjoint vertex sets it is equally easy to show that they are disjoint. $\square$

Since orientation is defined entirely in terms of vertices, there is no difficulty in transferring the concept of orientation from simplicial complexes to abstract simplicial complexes.

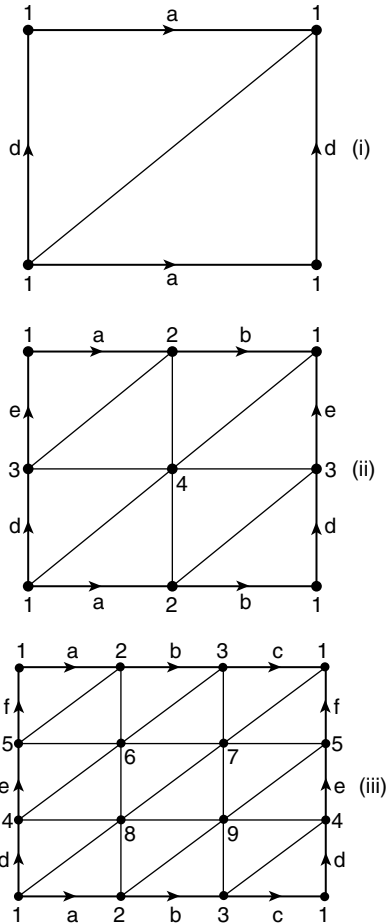
Triangulations and diagrams of simplicial complexes

A subset U of $\mathbb{R}^N$ which is the underlying space (3.14) of some simplicial complex is called a *polyhedron*. Of course there will almost certainly be many simplicial complexes whose underlying space is a given polyhedron U (the only exceptions are when U is finite); any of these is called a *triangulation* of U .

Something more will be said about triangulations in Chapter 5. For the present we are concerned with diagrammatic representation, which we shall discuss by means of an example which sufficiently illustrates what is involved, and in particular clarifies the meaning of diagrams such as those in 2.2(3), (4) and preceding 2.6.

If we cut a torus along two circles and open it out as in 2.2(3) the result is a plane rectangle with opposite sides similarly marked in order to remind us that they came from the same place on the torus. Consider the three diagrams below (which occurred in Chapter 2). We want, if possible, to regard these as recipes or 'schemas' for the construction of simplicial complexes in which dots, edges and triangles represent 0-, 1- and 2-simplexes respectively. To do this,

we first regard each of them as a schema for constructing an abstract simplicial complex $X = (V, S)$.



First label the vertices (dots) consistently, so that for example in any one diagram the beginning of a has the same label wherever it occurs. For notational convenience they have been labelled with numerals. Then take V to be the set of vertices, so that $V = \{1\}$ in (i), $V = \{1, 2, 3, 4\}$ in (ii) and $V = \{1, \dots, 9\}$ in (iii).

Next define S as follows: a subset $s \subset V$ belongs to S if and only if

- $s \in \{v\}$ for some $v \in V$
- or s is the set of vertices of an edge in the diagram
- or s is the set of vertices of a triangle in the diagram.

This defines an abstract simplicial complex $X = (V, S)$, which by 3.19 has a realization, but there are two questions to ask.

Qu. 1. Do edges and triangles in the diagram represent 1- and 2-simplexes of X ?

Qu. 2. What connexion, if any, is there between a realization of X and the torus from which we started?

3.20. The answer to Qu. 1 is 'yes' if and only if any given edge in the diagram has differently labelled end-points (which implies that any given triangle has three differently labelled vertices).

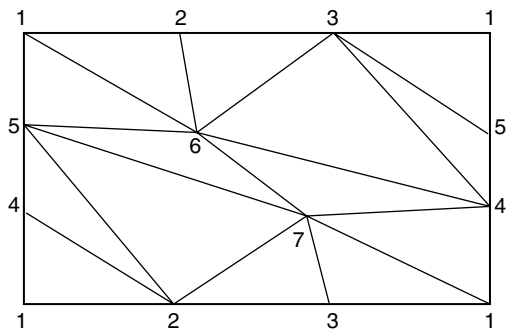
This is clear from the definition of an n -simplex of X as a set of vertices with precisely $n + 1$ elements. Thus (i) does not define an abstract simplicial complex of the kind required. Each edge and each triangle of (i) gives only a 0-simplex of X , and a realization of X is just one point. On the other hand for (ii) and (iii) the answer to Qu. 1 is 'yes'. (In (ii), it does not matter that a and b have the same set of end-points; what matters is that the end-points of a are different and the end-points of b are different.)

As for Qu. 2, this demands a less formal answer. When (ii) or (iii) is realized two edges with the same vertices will automatically be realized in the same place, i.e. will give rise to the same 1-simplex. The question is: does the realization look anything like the original torus?

3.21. Suppose that two edges (or two triangles) of the diagram have the same vertices if and only if they came from the same position on the torus before it was cut open. Then we shall regard a realization of X as a triangulation of the torus.

This means that when X is realized only the necessary identifications take place, and no extra ones. (It follows that the underlying space of the realization is homeomorphic to the original torus.) When the X coming from (ii) is realized a great many extra identifications take place (for example a and b^{-1} are identified), and the reader may like to try drawing the result. [Hint: the bottom half of the diagram is a repetition of the top half, and can be discarded.]

Examples 3.22. (1X) Theorem 3.19 assures us that the X coming from (iii) can be realized in $\mathbb{R}^5$ (see also 2.16(5)) whereas the original torus was in $\mathbb{R}^3$. Can this X in fact be realized in $\mathbb{R}^3$? A harder question (for the answer see Gardner (1975, 1978)) is whether the diagram below, which also gives a torus, can be realized in $\mathbb{R}^3$.

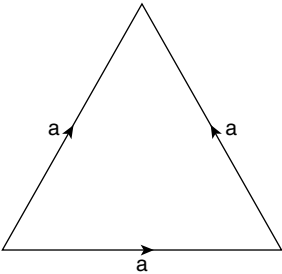


(2) In 2.6, (i) ensures that the answer to Qu. 1 is ‘yes’ while (ii) and (iii) relate to Qu. 2.



This diagram represents a Möbius band (compare 2.2(6)) if and only if there are at least five triangles. Reversing one of the arrows it represents a cylinder (obtained by identifying the ends of a rectangle without a twist) if and only if there are at least six triangles.

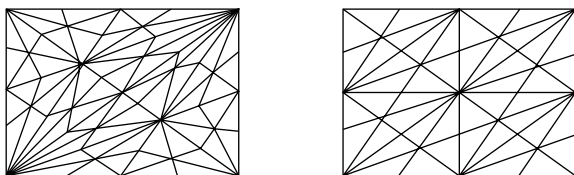
(4X) Find a triangulation of the object obtained by identifying all three sides of a triangle, as in the diagram (this is sometimes referred to as a *dunce hat*).



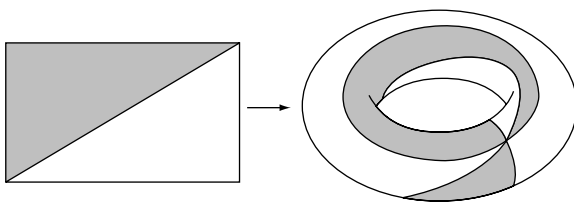
(5X) Cut a double torus (see 2.9) along suitable curves until it can be opened out flat, and hence find a triangulation. (You could use the method of 2.13 to check that your opened out diagram represents the double torus, which is orientable and has Euler characteristic -2 .)

(6) Any diagram of the type exemplified by (i)–(iii) above can be subdivided into one in which the answer to Qu. 1 (Page 80) is ‘Yes’ and in which the answer

to Qu. 2 is that the realization is homeomorphic to the torus (or whatever it was we started from). Two barycentric subdivisions are enough for (i), and one is enough for (ii). (Compare the discussion following 2.6.)



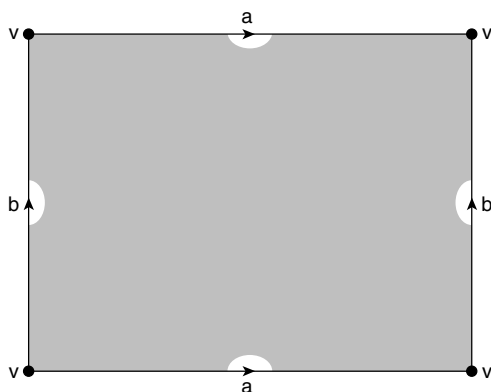
★(7) Diagrams such as (i)–(iii) above can be interpreted other than as recipes for the construction of abstract simplicial complexes. For example (i) can be interpreted as a covering of the torus by two triangles which are wrapped around the torus in such a way that each edge becomes a circle and all the



vertices land on the same point. Thus identifications take place on the boundaries of the triangles, but notice that there are no identifications of interior points. This kind of construction cannot be made with flat triangles, and the intersection condition 3.14(2) is hopelessly violated.

Likewise (ii) can be interpreted as eight triangles on the torus, and again the intersection condition is violated, though in this case each individual triangle lands on the torus without identifications. Of course with (iii) the intersection condition is satisfied.

These coverings by curved triangles which may have identifications on their boundaries and may violate the intersection condition are called *cell complexes*. It is possible to use much more general building blocks than triangles: for example the diagram below makes the torus into a cell complex with one 2-cell (the rectangle), two 1-cells (a and b) and one 0-cell (v).



This is certainly a desirable situation: calculations with cells are easier than with simplexes because there are fewer of them. But you never get something for nothing: the definition of cell complex is technically much more complicated than that of simplicial complex, and requires considerable preparation of a ‘general topology’ kind. One reason for this is that flat triangles can be defined simply by a little linear algebra, whereas for curvy cells one cannot get away from the concept of continuous mappings between subsets of, at least, euclidean spaces. For more information on cell complexes, see for example the books of Sato or Prasolov *Elements of Combinatorial and Differential Topology* listed in the References. ★

Stars, joins and links

In this section we shall introduce some standard terms used in connexion with simplicial complexes, and state a few results for the reader to verify. The section is intended primarily as a source of examples from which the reader may develop a greater facility in working with and thinking about simplicial complexes. Only 3.23, 3.29(2) and 3.29(5) are of significance later. The first two are straightforward while the third is a fairly hard exercise giving the construction of ‘stellar subdivisions’ which are used in Chapter 5, itself mostly ‘optional reading’. Enough is said in Chapter 5 to enable the reader who has not worked through 3.29(5) to follow the arguments, at any rate for the case of two-dimensional simplicial complexes.

Throughout, K is a simplicial complex.

3.23. Closure Let Σ be a set of simplexes of K . The *closure* of Σ (in K) is the smallest subcomplex of K which contains Σ , i.e. it is defined by

$$\overline{\Sigma} = \{s \in K : s < t \text{ for some } t \in \Sigma\}.$$

Other notations are $\text{Cl}(\Sigma)$ and $\text{Cl}(\Sigma, K)$. If $s \in K$ and $\Sigma = \{s\}$, then we shall write $\bar{s}$ rather than $\overline{\{s\}}$ for the closure of Σ . (Compare 3.15(5).) Note that $|\bar{s}| = s$ (3.14); indeed $|\Sigma| = |\bar{\Sigma}|$ for any Σ .

3.24. Star Let Σ be a set of simplexes of K . The *star* of Σ (in K) is defined by

$$\text{St}(\Sigma) = \text{St}(\Sigma, K) = \{s \in K : s > t \text{ for some } t \in \Sigma\}.$$

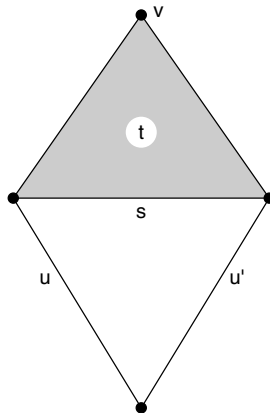
The *closed star* of Σ (in K) is the closure of the star of Σ , and is denoted $\overline{\text{St}}(\Sigma)$ or $\overline{\text{St}}(\Sigma, K)$. Thus

$$\begin{aligned} \overline{\text{St}}(\Sigma) = \{u \in K : \exists t \in \Sigma \text{ and } s \in K \\ \text{with } s > u \text{ and } s > t\} \end{aligned}$$

and $\overline{\text{St}}(\Sigma)$ is a subcomplex of K . The most important example is $\Sigma = \{s\}$: we write $\text{St}(s)$ and $\overline{\text{St}}(s)$ rather than $\text{St}(\{s\})$ and $\overline{\text{St}}(\{s\})$.

Examples 3.25. (1X) Let s_0 be a 0-simplex of K . Then $s_0 \in \text{St}(\Sigma) \Leftrightarrow s_0 \in \Sigma$. It follows from this that $\text{St}(\Sigma)$ is a subcomplex of K if and only if, for each component K_1 of K , Σ contains either all or none of the 0-simplexes of K_1 .

(2) Let K be the simplicial complex in the diagram, and let $\Sigma = \{s\}$. Then $\overline{\text{St}}(s) = \bar{t}$ where t is the 2-simplex. On the other hand $\text{St}(\bar{s}) = (\bar{t} \setminus \{v\}) \cup \{u, u'\}$. Thus $\overline{\text{St}}(\Sigma)$ and $\text{St}(\bar{\Sigma})$ may be sets neither of which contains the other.



(3X) For $s \in K$,

$$t \in \overline{\text{St}}(s) \Leftrightarrow s \text{ and } t \text{ are both faces of the same simplex of } K.$$

Thus the relation between s and t defined by $t \in \overline{\text{St}}(s)$ is symmetric (and reflexive). Is it transitive?

(4X) If $\Sigma_1 \subset \Sigma_2$ then $\text{St}(\Sigma_1) \subset \text{St}(\Sigma_2)$. Is the converse true?

3.26. Join Let $s_p = (v^0 \dots v^p)$ and $s_q = (w^0 \dots w^q)$ be simplexes in $\mathbb{R}^N$. They are called *joinable* if the $p + q + 2$ vectors $v^0, \dots, v^p, w^0, \dots, w^q$ are affinely independent, and in that case their *join* $s_p s_q$ is the $p + q + 1$ -simplex $(v^0 \dots v^p w^0 \dots w^q)$. (If they are joinable then their join is undefined.) Thus $s_p s_q = s_q s_p$ when either side is defined.

Two simplicial complexes K and L in $\mathbb{R}^N$ are called *joinable* if, for each $s \in K$ and $t \in L$, s and t are joinable, and if any two such joins, st and $s't'$, meet, if at all, in a common face of st and $s't'$. The *join* KL of K and L is then the simplicial complex

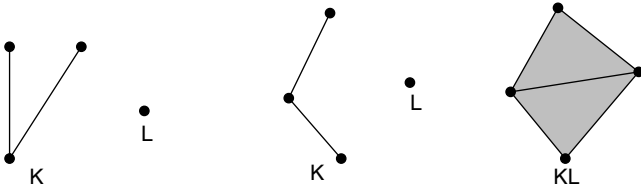
$$KL = K \cup L \cup \{st : s \in K \text{ and } t \in L\}.$$

Examples 3.27. (1XX) Let $a \in s_p$ and $b \in s_q$; then the segment joining a and b consists of points $\lambda a + (1 - \lambda)b$ for $0 \leq \lambda \leq 1$. We have: s_p and s_q are joinable if and only if any two such segments intersect at most in an end-point, and in that case

$$s_p s_q = \{\lambda a + (1 - \lambda)b : a \in s_p, \quad b \in s_q, \quad 0 \leq \lambda \leq 1\}.$$

Likewise two simplicial complexes K and L are joinable if and only if the segments $\{\lambda a + (1 - \lambda)b : 0 \leq \lambda \leq 1\}$ are disjoint except for end-points for all $a \in |K|$ and $b \in |L|$, and $|KL|$ is then the union of all such segments. It follows that if $|K| = |K_1|$ and $|L| = |L_1|$ and K and L are joinable, then so are K_1 and L_1 and $|KL| = |K_1 L_1|$.

(2) The simplicial complexes (in $\mathbb{R}^2$) on the left are not joinable, whereas those in the centre are, and their join is drawn on the right.



(3) Any K and the empty simplicial complex $\emptyset$ are joinable, and $K\emptyset = K$.

(4) If L consists of a single 0-simplex (v) , and K and L are joinable, then KL is also called the *cone* on K with vertex v . We shall meet this idea again in 6.8.

(5X) Writing α_p for the number of p -simplexes of a simplicial complex, and supposing KL is defined,

$$\alpha_p(KL) = \alpha_p(K) + \left(\sum_{q=0}^{p-1} \alpha_{p-q-1}(K) \alpha_q(L) \right) + \alpha_p(L)$$

(the middle sum being absent when $p = 0$). In particular let K and L be 0-dimensional, consisting of m and n 0-simplexes respectively. Then KL is a graph with cyclomatic number $(m-1)(n-1)$. (Clearly KL is bipartite (1.21(5)).)

(6X) If the simplexes in 3.26 are regarded as oriented, then the orientation of $\sigma_p \sigma_q$ can be defined by the ordering $(v^0 \dots v^p w^0 \dots w^q)$. In that case,

$$\sigma_p \sigma_q = (-1)^{(p+1)(q+1)} \sigma_q \sigma_p.$$

3.28. Link Let $s \in K$. The *link* of s in K is defined by

$$\text{Lk}(s) = \text{Lk}(s, K) = \{t \in K : st \in K\}.$$

Recall that $st \in K$ means (i) s and t are joinable, and (ii) the join st belongs to K . It follows that $\text{Lk}(s)$ is a subcomplex of K .

Examples 3.29. (1) In the case of the simplicial complex of 3.25(2), $\text{Lk}(t) = \emptyset$, $\text{Lk}(s) = \{v\}$, $\text{Lk}(v) = \bar{s}$.

(2X) Let K be a simplicial complex of dimension 2. Then the link of a 0-simplex v^0 consists of all the 0-simplexes v^1 for which $(v^0 v^1) \in K$ and all the 1-simplexes $(v^1 v^2)$ for which $(v^0 v^1 v^2) \in K$. The link of a 1-simplex $(v^0 v^1)$ consists of all 0-simplexes v^2 for which $(v^0 v^1 v^2) \in K$.

If we suppose that the link of every 0-simplex is a simple closed polygon, then it follows that the link of every 1-simplex is two 0-simplexes and that K is pure (every simplex is a face of a 2-simplex). Thus K consists precisely of the triangles of K together with their edges and vertices. Hence the definition of a closed surface given in 2.1 can be rewritten:

A closed surface is a connected two-dimensional simplicial complex in which the link of every 0-simplex is a simple closed polygon.

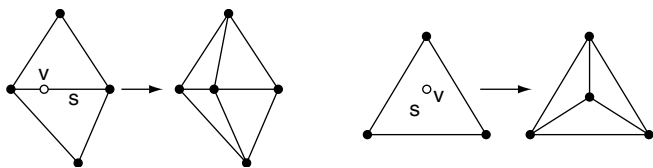
(3X) Let s_0 be a 0-simplex of K . Then

$$\text{Lk}(s_0) = \overline{\text{St}(s_0)} \setminus \text{St}(s_0).$$

Is this ever true of an n -simplex for $n > 0$?

(4X) Let $s \in K$. Then $\bar{s}$ and $\text{Lk}(s)$ are joinable, and $\bar{\text{Lk}}(s) = \overline{\text{St}}(s)$. In particular if s is a 0-simplex s_0 , then $\overline{\text{St}}(s_0)$ is the cone on $\text{Lk}(s_0)$ with vertex s_0 .

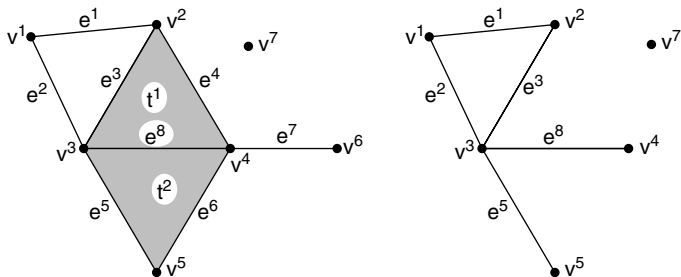
(5XX) Let $s \in K$ and let v be an interior point of s , i.e. a point of s not on a proper face of s . Let v also denote the simplicial complex whose only simplex is (v) . Then v and $\dot{s}$ (the set of proper faces of s , 3.15(5)) are joinable, and their join $v\dot{s}$ is joinable to $\text{Lk}(s, K)$. Further, $|v\dot{s}| = s$, and $|(v\dot{s})\text{Lk}(s, K)| = |\overline{\text{St}}(s, K)|$. (3.8(5) is relevant to the proof of $|v\dot{s}| = s$; see also 3.27(1).) Finally, $(K \setminus \overline{\text{St}}(s, K)) \cup (v\dot{s})\text{Lk}(s, K)$ is a simplicial complex with the same underlying space as K . It is said to be obtained from K by *starring s at v* . Repeated application of this procedure yields a *stellar subdivision* of K . Some examples of starring are drawn below.



Collapsing

The technique of collapsing is a simple geometrical idea which will be exploited in Chapter 6. Since, however, it does not assume any of the theory of homology groups, we shall introduce the idea here and give one or two mild applications.

Definitions 3.30. Let K be a simplicial complex. A *principal simplex* of K is a simplex of K which is not a proper face of any simplex of K . For any $s \in K$, a *free face* of s is a proper face t of s such that t is not a proper face of any simplex of K besides s . (Thus a free face of s has dimension $\dim s - 1$ and if s has a free face then s must be principal.)



In the left-hand diagram, the principal simplexes are:

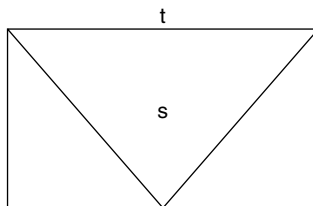
$$v^7; e^1, e^2, e^7; t^1, t^2.$$

Of these, only e^7, t^1, t^2 have free faces, namely v^6 for e^7 , e^3 and e^4 for t^1 , and e^5 and e^6 for t^2 .

Suppose that K has a principal simplex s which has a free face t . Then the subset K_1 of K obtained by removing s and t is a subcomplex of K , and is said to be obtained from K by an *elementary collapse* (which we describe as being a *collapse of s across t*). The notation is $K \searrow^e K_1$. The right-hand diagram above shows the result of three elementary collapses (t^1 across e^4 , t^2 across e^6 and e^7 across v^6); two further elementary collapses, of e^5 across v^5 and e^8 across v^4 , now become possible. A sequence $K \searrow^e K_1 \searrow^e K_2 \searrow^e \dots \searrow^e K_m$ of elementary collapses is called a *collapse*, and is denoted by $K \searrow K_m$. (A special case is the empty sequence by which $K \searrow K$!) If K_m consists of a single vertex we say that K *collapses to a point*.

This concept has already been encountered for graphs in 1.12(2): a graph collapses to a point if and only if it is a tree.

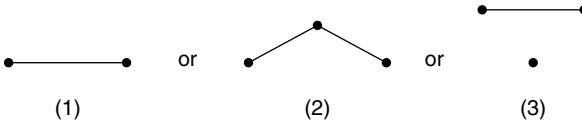
Let $\mathcal{D}$ be the plane region (a *disk*) enclosed by a simple closed polygon (see 1.25), and let D be a triangulation of $\mathcal{D}$, i.e. a simplicial complex (of dimension two) such that $|D| = \mathcal{D}$. Our object is to prove that D collapses to a point. A naïve argument would run: ‘Collapse one triangle of D across a free face on the boundary polygon. What is left is a triangulation D' of a region $\mathcal{D}'$ enclosed by a simple closed polygon, and by induction $D' \searrow$ point, Q.E.D.’ Unfortunately, this will not do, as can be seen by collapsing s across t in the diagram below: what is left is not enclosed by a simple closed polygon. The trouble is that s meets the boundary polygon in an edge and an isolated vertex.



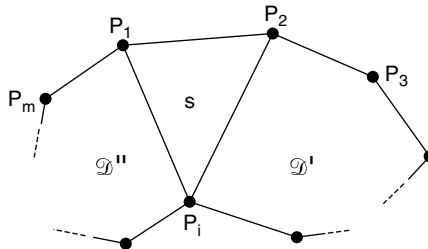
Lemma 3.31. *Let us denote the simple closed polygon enclosing $\mathcal{D}$ by ∂D . Suppose that D contains at least two triangles; then D contains at least two triangles meeting ∂D in an arc, i.e. in an edge or two edges. (See 3.22(3) for*

a picture which, on removing the arrows, gives an example where D contains exactly two such triangles.)

Proof The proof is by induction on $\alpha_2(D)$ the number of triangles of D : clearly it is true when $\alpha_2 = 2$ so suppose it holds when $\alpha_2 < n$ ($n \geq 3$) and let $\alpha_2(D) = n$. Let s be a triangle of D meeting ∂D in at least an edge – clearly D must contain at least two such triangles. Then $s \cap \partial D$ is



If there are no triangles of type (3) in D then all choices of s meet ∂D in an arc, so the result is proved. If there is a triangle s of type (3), proceed as follows (see the diagram below). Label the vertices of ∂D by $P_1, P_2, P_3, \dots, P_m$ where s has vertices $P_1 P_2 P_i$ for some i , $4 \leq i \leq m - 1$. Then $P_2 P_3 \dots P_i P_2$ and $P_1 P_i P_{i+1} \dots P_m P_1$ are simple closed polygons enclosing plane regions $\mathcal{D}'$, $\mathcal{D}'' = 1$ triangulated by D', D'' , say. If $\alpha_2(D') = \alpha_2(D'') = 1$, then the unique triangles in D', D'' meet ∂D in an arc. If $\alpha_2(D') > 1, \alpha_2(D'') = 1$, then by the induction hypothesis choose two triangles of D' meeting $\partial D'$ in an arc. At most one of these can contain the edge $P_2 P_i$; the other[†], together with the unique triangle in D'' , both meet ∂D in an arc. The other cases are treated similarly; in every case D contains at least two triangles meeting ∂D in an arc.

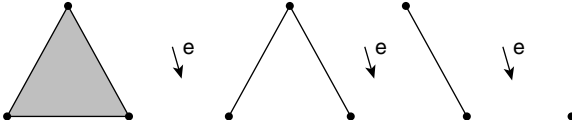


□

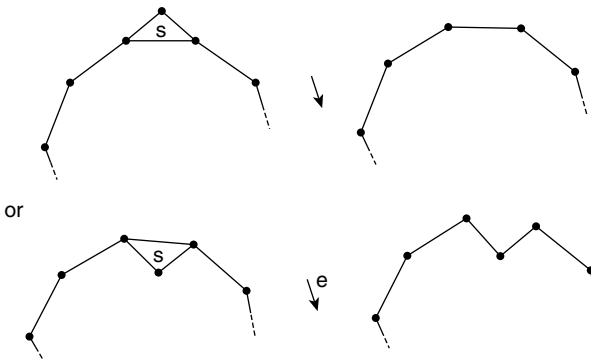
Theorem 3.32. *With the above notation, any triangulation D of $\mathcal{D}$ collapses to a point.*

[†] Notice that the proof would break down at this point if we were trying to prove the weaker assertion that D always contains at least one triangle meeting ∂D in an arc. This is a not uncommon feature of proofs by induction.

Proof Clearly the theorem holds when $\alpha_2(D) = 1$, the collapse then being



Suppose the theorem is true when $\alpha_2(D) < n$ ($n \geq 2$) and let $\alpha_2(D) = n$. By the lemma choose a triangle s of D meeting ∂D in an arc, and collapse as follows:



In either case $D \searrow D_1$ where D_1 is a triangulation of the plane region bounded by a simple closed polygon, and $\alpha_2(D_1) = n - 1$. Thus $D \searrow D_1 \searrow$ point, by the induction hypothesis. $\square$

The Euler characteristic of a simplicial complex K of dimension n is $\chi(K) = \alpha_0(K) - \alpha_1(K) + \alpha_2(K) - \dots + (-1)^n \alpha_n(K)$, α_i being the number of simplexes of dimension i in K , $i = 0, \dots, n$. (Compare 2.11.) Clearly if $K \searrow L$ then $\chi(K) = \chi(L)$ (consider the effect of an elementary collapse). Hence:

Corollary 3.33. *If D is a triangulation of the plane region $\mathcal{D}$ enclosed by a simple closed polygon, then $\chi(D) = 1$. (Compare the proof of 2.13.)* $\square$

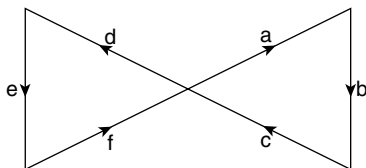
There is another consequence of 3.31 which will be useful later.

Theorem 3.34. *Let D be a triangulation of the plane region $\mathcal{D}$ enclosed by a simple closed polygon ∂D , and let the simplicial complex K be obtained from D by deleting one 2-simplex. Then $K \searrow \partial D$.*

Proof There is nothing to prove if $\alpha_2(D) = 1$, for then $K = \partial D$ and the empty sequence of elementary collapses will work! Suppose the result is true

when $\alpha_2 < n$ ($n \geq 2$) and that $\alpha_2(D) = n$. By the lemma (3.31) K has at least one triangle s meeting ∂D in an arc. Collapse as in the proof of 3.32; let this take D to D' and K to K' . Then D' is a triangulation of a disk containing $n - 1$ triangles, and K' is obtained from D' by removing a single triangle s ; hence by the induction hypothesis $K' \searrow \partial D'$. Now none of the elementary collapses comprising this collapse can be across a face of s , since the face or faces of s in K' are still present at the end, in $\partial D'$. Hence $K = K' \cup \bar{s} \searrow \partial D' \cup \bar{s}$ where $\bar{s}$ denotes, as usual, the set of all faces of s including s itself. Since $\partial D' \cup \bar{s} \searrow \partial D$ this completes the induction. $\square$

Examples 3.35. (1) This result can be used to give an alternative proof of 2.13 without using $\chi(D) = 1$. For removing a single triangle from a closed surface gives a simplicial complex which collapses to the one-dimensional subcomplex which appears, in a polygonal representation of the surface in the plane, as the boundary of the polygon. The Euler characteristic of this graph is easy to calculate, and the removal of a triangle at the beginning just lowers the Euler characteristic by 1. For example the triangulation of a torus in 2.2(3) collapses, after removal of a triangle, to the graph below which has Euler characteristic -1 . Thus the original simplicial complex has Euler characteristic 0.



(2) The triangulations of a cylinder and a Möbius band in 3.22(3) collapse to simple closed polygons (sometimes called *spines* of the original simplicial complexes).

(3X) Let M be a closed surface and let K be a subcomplex of M which does not contain all the triangles of M . Supposing that K contains at least one triangle, then it contains at least one triangle with a free face, and therefore K collapses to a subcomplex with fewer triangles and so by induction to a graph. (To see that K must have a triangle with a free face, use 2.2(5) to show that otherwise K would have to contain all the triangles of M .)

(4) It is perhaps worth noting that 3.32 does not generalize to higher dimensions. A simplicial complex in $\mathbb{R}^3$ whose underlying space is homeomorphic to the unit sphere $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ in $\mathbb{R}^3$ will enclose a region known as a *3-ball*. It is possible to find such a simplicial complex and a

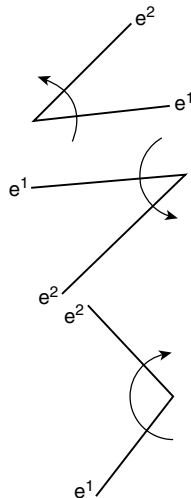
triangulation of the resulting 3-ball which does not collapse to a point. A famous example is called ‘Bing’s house with two rooms’. See Bing (1964).

★ Appendix on orientation

The passage from simplexes to oriented simplexes is an essential one for setting up the homology theory described in this book. The concept of orientation arises also in the study of real vector spaces, and in this appendix we shall relate these two ideas as well as proving the basic property of orientations – that there are always exactly two of them.

Let V be a real vector space of dimension $n \geq 1$ provided with a definite basis, called the *preferred basis*, and suppose that $e^1, \dots, e^n$ and $f^1, \dots, f^n$ are *ordered bases* of V , that is they are bases in which the vectors come in a prescribed order. (Such a basis is sometimes called an *n-frame*.) We want to compare the ‘orientations’ of these two bases.

Take for example $V = \mathbb{R}^2$. Then geometrically e^1, e^2 and f^1, f^2 have the same orientation if and only if they are both ‘clockwise’ or both ‘anticlockwise’ – more precisely this means that the rotations from e^1 to e^2 and from f^1 to f^2 , in each case through an angle $< 180^\circ$, are both in the same direction. Thus in the diagram below the first and second bases have the same orientation, not shared by the third basis. When $V = \mathbb{R}^3$ then geometrically we require that the two bases should both be ‘right-handed’ or both ‘left-handed’. Thus for example for any basis e^1, e^2, e^3 and e^2, e^1, e^3 have opposite orientation while e^1, e^2, e^3 and $-e^2, e^1, e^3$ have the same orientation.



Now we ask the question: can we find a continuous family[†] of ordered bases $u^1(t), \dots, u^n(t)$, depending on a parameter $t \in [0, 1]$, such that for $t = 0$ this basis is $e^1, \dots, e^n$ and for $t = 1$ it is $f^1, \dots, f^n$? (The important thing is that for every $t \in [0, 1]$ we must get a basis; we can turn e^1, e^2 (an ordered basis of $\mathbb{R}^2$) into e^2, e^1 by the family $(1-t)e^1 + te^2, te^1 + (1-t)e^2$, but for $t = \frac{1}{2}$ these vectors are not linearly independent.) The reader can probably convince himself that for $V = \mathbb{R}^2$ and $V = \mathbb{R}^3$ such a family exists if and only if the two ordered bases have the same orientation in the intuitive sense described above.

Any linear isomorphism $V \rightarrow V$ determines, relative to the preferred basis in each copy of V , a nonsingular real $n \times n$ matrix. The isomorphism is called *orientation preserving* if the determinant of this matrix is positive, and *orientation reversing* if it is negative. Note that this definition does not in reality depend on the choice of preferred basis, for if A is the matrix relative to the preferred basis then the matrix relative to any other basis is of the form $P^{-1}AP$ for some nonsingular P , and this has the same determinant as A . I now claim the following.

Theorem 3.36. *Let $e^1, \dots, e^n$ and $f^1, \dots, f^n$ be ordered bases of V . The following statements are equivalent:*

- (i) *There exists a continuous family of ordered bases taking $e^1, \dots, e^n$ to $f^1, \dots, f^n$.*
- (ii) *The unique isomorphism $V \rightarrow V$ taking e^i to f^i for $i = 1, \dots, n$ is orientation preserving.*

When these hold we say that the two bases have the same orientation, otherwise that they have opposite orientation.

The theorem will be proved at the end of this appendix. It says that ordered bases can be divided into two orientation classes in such a way that two bases in the same class can be connected by a continuous family and two bases in different classes cannot. The classes can be distinguished by means of a determinant.

Examples 3.37. (1X) $e^1, \dots, e^n$ and $e^{\pi(1)}, \dots, e^{\pi(n)}$, where π is a permutation of the set $\{1, \dots, n\}$ have the same orientation if and only if π is an even permutation. It is enough to check that if π is a transposition then the two bases

[†] Write $u^j(t) = (a_{1j}(t), \dots, a_{nj}(t))$, where these are the coordinates of $u^j(t)$ relative to the preferred basis. Then we require that the n^2 functions $a_{ij}(t)$ should be continuous. (This definition is in fact independent of the choice of preferred basis.)

have opposite orientation, and this is most easily proved by taking (as we may) $e^1, \dots, e^n$ itself to be the preferred basis.

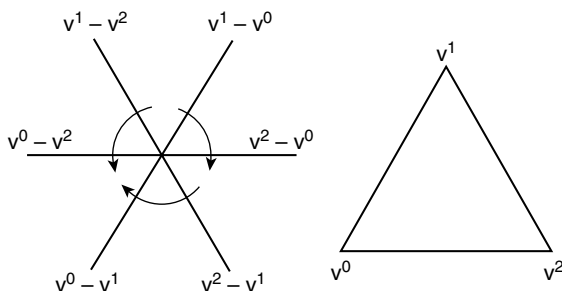
(2X) $e^1, \dots, e^n$ and $-e^1, \dots, -e^n$ have the same orientation if and only if n is even.

Now let us turn to *orientation of simplexes*. Let $(v^0 \dots v^n)$ and $(v^{\pi(0)} \dots v^{\pi(n)})$ be two ordered simplexes, where π is a permutation of $\{0, \dots, n\}$. The vectors $v^0, \dots, v^n$ lie in an affine subspace U^A of dimension n in $\mathbb{R}^N$, while the vectors $v^i - v^j$ lie in a linear space U of dimension n parallel to U^A . Consider the following two bases of U .

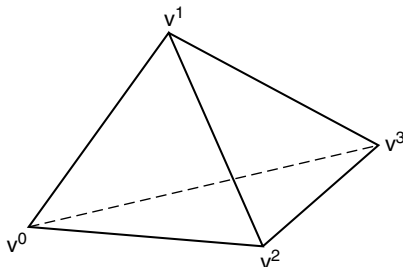
$$v^1 - v^0, \dots, v^n - v^0$$

$$\text{and } v^{\pi(1)} - v^{\pi(0)}, \dots, v^{\pi(n)} - v^{\pi(0)}.$$

Definition 3.38. $(v^0 \dots v^n)$ and $(v^{\pi(0)} \dots v^{\pi(n)})$ have the *same* or *opposite orientation* according as these two bases of U have the same or opposite orientation.



Examples 3.39. $(v^0 v^1 v^2) = (v^1 v^2 v^0)$ (same orientation)
 $(v^0 v^1 v^2) = -(v^2 v^1 v^0)$ (opposite orientation)



$v^1 - v^0, v^2 - v^0, v^3 - v^0$ is 'right-handed'; $v^3 - v^2, v^1 - v^2, v^0 - v^2$ is 'left-handed'.

Hence $(v^0 v^1 v^2 v^3) = -(v^2 v^3 v^1 v^0)$.

Proposition 3.40. $(v^0 \dots v^n)$ and $(v^{\pi(0)} \dots v^{\pi(n)})$ have the same orientation if and only if π is an even permutation.

Proof It is enough to check that if π is a transposition then the two simplexes have opposite orientation. Assuming that π is a transposition leaving 0 fixed the result follows from 3.37(1). Assuming on the other hand that π transposes 0 and i for some $1 \leq i \leq n$, the matrix, referred to the first basis above, of the isomorphism taking the first basis to the second, has the following form.

$$\begin{pmatrix} 1 & & & & & & & & \\ & 1 & & & & & & & \\ & & \ddots & & & & & & \\ & & & 1 & & & & & \\ -1 & -1 & \dots & -1 & -1 & -1 & \dots & -1 & \\ & & & & & 1 & & & \\ & & & & & & \ddots & & \\ & & & & & & & 1 & \end{pmatrix}$$

Here the blank entries are all zero, and the line of -1 s is the i th row. This clearly has determinant -1 . $\square$

To conclude this appendix, here is a proof of Theorem 3.36.

Proof of 3.36 Any basis of V determines a (real) nonsingular $n \times n$ matrix whose j th column consists of the coordinates of the j th basis vector relative to the preferred basis. If E and F are the matrices determined in this way by the given bases, then the matrix, relative to the preferred basis, of the unique isomorphism taking e^i to f^i for $i = 1, \dots, n$ is FE^{-1} . Thus statements (i) and (ii) of the theorem are respectively equivalent to

(i)' there is a continuous family of nonsingular $n \times n$ matrices taking E to F ; and (ii)' the determinants $|E|$ and $|F|$ have the same sign.

The implication (i)' $\Rightarrow$ (ii)' is a straightforward consequence of the fact that the determinant of a matrix, being a polynomial in the entries of the matrix, is a continuous function of the entries. Thus if the general matrix of the family is A_t (so that $A_0 = E$ and $A_1 = F$) then $|A_t|$, being continuous and non-zero for $0 \leq t \leq 1$, must have the same sign for $t = 0$ and $t = 1$.

To prove (ii)' $\Rightarrow$ (i)' we must exhibit a continuous family taking E to F . It is enough to construct a continuous family taking E to one of the two diagonal matrices $\text{diag}(1, 1, \dots, 1, \pm 1)$. The family we shall construct will be made up of a sequence of families, corresponding to a sequence of operations which reduce E to one of these two matrices. Let A_0 be any nonsingular $n \times n$ matrix; consider the following two operations.

- I. Multiply all the elements of one row (or column) of A_0 by $\lambda > 0$.
- II. Add to one row (or column) of A_0 any multiple μ of a different row (or, respectively, column).

Note that neither of these affects the sign of $|A_0|$. Suppose A_1 is obtained from A_0 by an operation I. Then the general member of a continuous family taking A_0 to A_1 is A_t , where A_t is obtained from A_0 by an operation I, replacing λ by $1 - t + \lambda t$. Similarly an operation II can be achieved via a continuous family.

Now using operations I and II any nonsingular $n \times n$ matrix $E = (a_{ij})$ can be reduced to a diagonal matrix with 1's and -1 's on the diagonal. We show how to turn E into a matrix of the form

$$\begin{pmatrix} \pm 1 & 0 & \dots & 0 \\ 0 & \boxed{} \\ \vdots & & & \\ 0 & & & \end{pmatrix}$$

where the box contains an $(n-1) \times (n-1)$ nonsingular matrix; the proof then proceeds by induction. In fact, if $a_{11} = 0$ then choose a non-zero entry in the first row or column (they can't all be zero); say this is the entry a_{i1} in the i th row. Then by an operation II, with $\mu = 1/a_{i1}$, we obtain a matrix with top left-hand entry 1. If $a_{11} \neq 0$ multiply the first row by $1/|a_{11}|$ (an operation I), obtaining a matrix with top left-hand entry ± 1 . Now the other entries in the first row and then the first column can easily be made zero by operations II.

Next, two -1 's on the diagonal can be turned into $+1$'s:

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & -1 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Each operation is of type II. Similarly a 1 and a -1 can be interchanged by a sequence of operations of type II. This completes the proof that every E can be reduced to one of $\text{diag}(1, 1, \dots, 1, \pm 1)$. $\square$

An alternative statement of the theorem is that the general linear group $GL(n, \mathbb{R})$ contains precisely two path components for any $n \geq 1$. The proof above constructs a polygonal path joining any two matrices in this group with determinants of equal sign, regarding $GL(n, \mathbb{R})$ as being contained in $\mathbb{R}^{n^2}$ in the natural way. ★